

VITALIY PIKALO

412-314-1013 • vap5288@psu.edu • vitaliypikalo.com • github.com/Vitaliy-Pikalo • Pittsburgh, PA

EDUCATION

The Pennsylvania State University, Erie — Behrend College

Erie, PA

B.S. Computer Engineering GPA: 3.25

Expected: Dec 2027

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Web), Digital Systems & Design Lab, Circuits & Devices, Matrices (Linear Algebra), Differential Equations, Calculus I-II, Wave Motion & Quantum Physics, Statistics

SKILLS

Languages: C++, Python, Java, JavaScript

Frameworks & Tools: Git, Flutter, Firebase, Supabase, Stripe, scikit-learn, pandas, SQLite, TensorFlow, OpenCV, HTML/CSS, REST APIs

Languages (spoken): Ukrainian (native), Russian (native), Spanish (intermediate)

EXPERIENCE

Full-Stack Development Team Lead — Freelance / Client Work

2025

- Led a 4-engineer team to ship **10+** full-stack web and mobile booking apps for local businesses, processing **100+** customer reservations per month
- Cut checkout friction by building encrypted Stripe payment flows with Supabase auth, keeping transactions PCI-compliant across all clients
- Ran branching, PR review, and frontend/backend workflows on GitHub across concurrent client projects

Equity Analyst — Intrieri Family Student Managed Fund, Penn State Behrend

2025

- Co-managed a **\$1.6M** live equity portfolio with a team of analysts under CFA Institute professional and ethical standards
- Produced economic, industry, and equity-valuation analyses that fed directly into the fund's buy/sell decisions and portfolio rebalancing

AI Teaching Assistant & Guest Lecturer — Penn State

2026

- Delivered guest lectures to **100+** students on prompt engineering, AI workflows, and automation with Microsoft Copilot, invited by faculty
- Built instructional materials and ran 1:1 sessions, raising assignment completion quality

PROJECTS

Chess960-NN — AlphaZero-Style Engine [live demo](#) · [github](#)

2026

PyTorch, FastAPI, MCTS, reinforcement learning, python-chess, Stockfish

- Built and **deployed** an AlphaZero-style Chess960 engine — PyTorch 10-block ResNet (**6.7M** params) + PUCT MCTS — served live via a FastAPI backend (Hugging Face Spaces + Netlify) to play it or watch it face Stockfish
- Trained from scratch on a single RTX 3060 Ti: supervised pretraining on **6.4M** positions, then RL self-play with a gated promotion loop; benchmarked to **~1600 Elo** over 100 games vs Stockfish

Kalshi Trading Bot github.com/Vitaliy-Pikalo/kalshi-trading-bot

2026

Python, SQLite, REST APIs, scikit-learn, daemon architecture

- Engineered a fault-tolerant 24/7 data pipeline + daemon ingesting **270K+** live market snapshots into SQLite, surviving restarts with the DB as source of truth
- Fixed concurrency failures with WAL mode + busy-timeout and resolved an **~18x** data-inflation join bug; ran versioned A/B experiments to validate behavior

Bonfire — Multiplayer Party Game

2026

Flutter, Firebase / Firestore, Riverpod, Cloud Functions, Groq LLM

- Built core gameplay (rooms, turns, voting) on a 3-developer team using Flutter + Firestore with Riverpod state management and typed repositories
- Integrated a Groq LLM through Firebase Cloud Functions and enforced Firestore security rules with

serialization round-trip tests

LEADERSHIP, HONORS & ACTIVITIES

- **Varsity Men's Tennis (NCAA D-III):** AMCC Player of the Week & All-Tournament Team; 17-0 in singles, led team to a 7-0 season, the 5th straight AMCC title, and an NCAA Division III tournament berth
- **Student Government Senator:** co-founded the campus AI taskforce
- **Chess & Robotics Clubs:** ran sponsored tournaments (\$100+ prizes); built a navigation rover
- **Honors:** Dean's List (Spring 2026)